



Four Banjo Necks and One Pot

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Four necks of different type of wood with the same shape and hardware were attached to a single pot, with the same set-up. Sound files of played music and single plucks are provided. Most people will recognize small differences in the pluck sounds and identify the necks (albeit not unambiguously) in the played music either from experience or “training” from labeled examples. It is not known whether the differences arise solely from the wood or from some other inadvertent variation. Spectrograms show differences in intensity and decay time across the entire frequency range. These differences reflect differences in timbre and its time evolution. No simple physics narrative is identified to connect weight and large-scale stiffness to the observed sound variations.

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I. INTRODUCTION

Banjo and guitar necks certainly flex and twist due to the force at the nut or fret of each vibrating string.[1] Hence, there is a force *on* the strings that differs from what it would be were the neck perfectly rigid. This, in turn, impacts how the strings vibrate and drive the head and then how the head drives the air. In addition, neck vibrations combine with the rim to produce “body modes” whose effects are subtle but also contribute to the sound. So, an important question of the neck’s impact on sound is one of degree. Are the differences big enough to be heard and big enough to distinguish between necks?

A narrower question of some practical interest is whether the wood species of the neck impacts the sound to an extent that can be heard and distinguished. To that end, four blanks, i.e., maple, mahogany, walnut, and white oak, were milled identically on the Deering Goodtime CNC production line, equipped with identical hardware, and then mounted sequentially, with identical set-up, on a single Goodtime maple pot.

The weight of the wood alone varies from 10 to 16 ounces. The common tuners, hanger bolts, and frets were all together an additional 5 ounces. The assembled banjos were about $4\frac{1}{4}$ lbs (which included a metal arm rest), making the variation of neck wood weight a noticeable feature. The large-scale stiffness of the maple is 40% greater than the others. So, the physical differences among the necks are substantial. Also, there is no truss rod and no separate fretboard, making wood species more relevant here than for instruments so equipped. (Heavier and stiffer mean less motion at the nut.)

Some banjo players know that there is a sound — albeit on average — of mahogany *versus* maple necks. Other players know that you cannot distinguish any neck wood variations just from the sound. Yet others are not so sure. Among wood species proponents with first-hand experience, some acknowledge that the distinctions are subtle and may be more evident to the player than the listener.

Strictly speaking, from a single sample of a given wood type, one cannot tell what the average would be of a large set or what the variation would be *within* that set. And, to the extent that the perceived sound differences between these four necks are small, it may be due to differences other than the wood itself. Examples might be the glued-in polycarbonate

nuts, the fret setting, the neck mounting, or the set-up. Nevertheless, the sounds here might be of interest, particularly with respect to preconceived notions.

II. OUTLINE

In the **INTRODUCTION**, it was asked, “Are the differences big enough to be heard and big enough to distinguish between necks?” So, the first order of business is to listen to examples. In §IIIA, there are plucks on the open 1st string. In §IIIB, there are picked and frailed music samples. The reader is offered both species-labeled and unlabeled versions and encouraged to try to identify which is which.

The hardware is described in §IVA. §IVB presents spectrograms of the single plucks which show the time dependence of the frequency components.

Attaching a capo at the 1st fret removes the issue of whether the nut slots are the same on the different necks. Capoing further up the neck, i.e., at the 5th, fret also allows a comparison where motion at that capo location is certainly less than with 1st fret capoing or full scale length. Maple and mahogany 1st string plucks are compared this way in §V. In addition to spectrograms, the envelopes of the whole pluck waveforms highlight the neck dependence in a different way.

Section §VI reports a very different kind of experiment that was performed on a Deering Eagle II, a 10 lb banjo with a maple neck, an ebony fretboard, and a two-way truss rod. Admittance measurements on the banjo and basic string physics are used to produce a physical synthesis of the sound. This procedure allows a comparison of the actual banjo with an idealized version having an absolutely rigid neck.

Section §VII discusses grain variation, and section §VIII gives the gross physical parameters of the four necks. Section §IX has a few comments on the privileged position of the player. And section §X emphasizes why this is not a simple physics problem.

III. LISTEN!

The linked sound files are of controlled “wire-break” plucks and brief music selections, both picked and frailed. In each context, the recording procedure was as identical as possible — to the extent afforded by a reasonable amount of care. Those procedures are described in detail in section §IV.

A. Single Plucks

First consider controlled plucks on the 1st string. There are twelve plucks in six pairs of two. The two are on the same neck but performed after the bridge, pick-up, flat-plate resonator, and banjo-on-its-stand were slightly moved and then repositioned according to the original protocol. The first pair is maple, labeled mpl-I. The second pair, mpl-II, is also maple but a different neck, manufactured six years earlier. The next pair is mahogany — mhg-I. The fourth pair, mhg-II, is the same mahogany neck but a few days later, after other necks were mounted in the interim. So, every detail of neck attachment and set-up might inadvertently be a bit different. The fifth pair is walnut — wal. And the last pair is white oak — wok. You can listen to all twelve in a row or as separate files for each pair in the folder

<http://www.its.caltech.edu/~politzer/neck-wood/plucks/>

The sounds are certainly similar and may seem indistinguishable. Thresholds for discrimination vary widely among people for a variety of reasons. In the present case, after listening for attack and overall sustain, it might help to pay attention to how the tone evolves in time. The spectrograms presented in section §IVB2 for this recording reveal that there certainly are differences between the necks. For some harmonics, it is different overall intensity. For others, it is a matter of sustain, and that gives different necks different timbre as a function of time. There remains an important, obvious question. Are the two different maple necks closer to each other than to any of the others? Similarly, does re-setting everything with the one mahogany neck result in sounds that are closer than they are to the other necks? Careful listening and attentive examination of the spectrograms might give answers, but those would be judgement calls. A proper mathematical, statistical analysis is easier to imagine than it would be to perform. (It would be routine AI, given hundreds or thousands banjos.)

B. Played Music

Recordings were made with each of the four new necks, about 40 seconds long, with metal picks and bare-finger frailed. All selections with a given neck were recorded sequentially, in the same session. No hardware was moved. Each ~40 second selection appears only

once, with a unique file name. The recordings with a given neck differ within the normal range of playing technique that occurs when a modest player attempts to repeat the same performance. Care was taken with subsequent necks that the recording set-up was identical.

There are picked and frailed recordings with files labeled with an obvious three-letter wood identification. These include notes on all strings, over the whole neck, played with a variety of attack. The same tunes are recorded again and presented in files identified with A, B, C, D, and E rather than the wood type. This offers an obvious possible test of whether each neck has its own characteristic sound. Can a listener match the A, B,... with mpl, mhg, wal, or wok? At least some identifications will likely be tentative or ambiguous. Note that there are five unidentified labels and four types of wood; one wood type appears twice in the unidentified set, albeit from a different recording.

There are several possible strategies for attempting this challenge oneself. One may try based on previous experience or preconceptions, or one may try to match to the species-labeled versions. Some people find restricting to the picked versions most efficient. Also, comparing short selections at a time, e.g., a measure or two, highlights the differences. One might choose to work through the whole ~ 40 seconds that way, going back and forth between different recordings. Alternatively, a streamlined version might be sufficient: to facilitate that approach, the first four beats of each picked recording is reproduced in files similarly labeled but including “3sec” in the file name.[2]

All are in the folder

<http://www.its.caltech.edu/~politzer/neck-wood/TUNES-mp3s>

An answer key is available elsewhere.[3] Whether someone can hear the differences or not depends on their hearing. Whether they care about the differences is up to them.

IV. RECORDING DETAILS & QUANTITATIVE ANALYSES

The word “sound” is used both to refer to people’s perceptions and to refer to physical vibrations, typically in the air but possibly in wood or steel, and more specifically longitudinal waves of compression. The relation between the two has some aspects that are complex and subtle and others that are quite surprising.[4] This is part of the challenge and interest of the physics of music.

A. hardware & procedures

Each neck kept its own set of strings to ease the task of swapping necks. Those strings were brand new and the same gauge (9, 10, 13, 20w, 9).

1. *single plucks*

“Wire breaks” produce very reproducible plucks. #42 copper wire (0.0025” diameter) was looped around the 1st string, $5\frac{1}{2}$ ” away from the bridge, and pulled sideways (perpendicular to the string, parallel to the head) until it broke. Rather than recording sound with a microphone, which depends strongly on microphone position and surrounding geometry, the motion of the bridge was recorded with a piezo pickup. The pickup was glued to the bridge so that it was in the same place on the bridge each time the banjo was reassembled. The bridge was located each time using the standard matching of 1st and 4th string 12th fret and its harmonic. (This is one of the several setup features that are sources of variability.)

All played banjos have a back of some sort — the player serving as the back of an open-back.[5] Since most discussions of neck wood species are among resonator banjo players, the Goodtime pot was equipped with a snap-on wooden disk back, spaced $\frac{1}{4}$ ” from the bottom of the rim. A wood back typically gives stronger high frequency response (i.e., above 4000 Hz) than a player’s belly.

The bridge pickup, shown in Fig. 1, was a kna BP-1. It comes with an 11” cable, which was tied to one of the hooks with a cable tie. The pickup, its cable, and its $\frac{1}{4}$ ” jack socket weigh 18 gm. The pickup signal was converted to USB, plugged into a Mac, and recorded and analyzed with Audacity.

An ideal piezo pickup is an accelerometer. The piezoelectric material, when pushed on one surface with the opposite surface free, is squeezed by the surface acceleration due to its inertia (as long as the forcing frequency is below the first piezo resonance in that direction). The produced voltage is proportional to the squeeze, and the squeeze is proportional to the acceleration. As a function of frequency, acceleration is one power of frequency stronger than velocity or admittance, and admittance is a logical surrogate for the produced sound. Admittance-based synthesis produced a convincing version of banjo sound, needing only a small boost with frequency (a fractional power) to match real sound.[6] Bridge acceleration



FIG. 1: bridge and nut piezo pickups

might err slightly in the other direction.

Fig. 1 also shows a kna UK-1 ukulele pickup, which was cut down to the width of the nut and secured in place, just beyond the nut, between the strings and a maple shim. This pickup monitored motion of the nut and could be recorded simultaneously with the bridge pickup (using two computers). The nut certainly moves when the banjo is played. Processed into a music sound sample, it is both like and unlike any normal banjo sound. The motion was different in its details for different necks, but those differences had no simple correlation with variations in bridge motion. For some frequencies and some necks, both bridge and nut moved more than for other necks. In other circumstances, larger nut motion accompanied smaller bridge motion. Shedding no light on the central issues at hand, the nut motion is not described further.

2. *sound recordings*

The sound recordings used an AKG C1000S mic, into a Sure X2U USB adapter, and into the same Mac. The room was not ideal for sound recording, but care was taken to not move anything between recordings.

The pickup-glued bridge was replaced by a normal Deering bridge, and the banjo was

played open-back.

The picked selection was played with metal finger picks, and the frailed selection was played with bare fingers and nails.

B. single pluck spectrograms

The displayed frequency range in Fig. 2, i.e., up to 18,000 Hz, may well be more than is relevant to most listeners. It was chosen to reveal variations between necks.

1. *wire-break repeatability*

Commercial magnet wire is quite uniform. The biggest variations from pluck to pluck presumably come from how each bit of wire is handled and how the break is performed. The most obvious variable aspects are the position and angles of the wire just before it breaks under the applied tension.

The spectra and spectrograms of wire-break plucks are essentially indistinguishable if nothing else is altered between plucks. Because the neck comparisons necessarily involved re-mounting the necks and re-doing the set-up, their impact was investigated by purposely putting each banjo out of adjustment and then back in and even remounting a given neck. The magnitude of these readjustment issues can be gauged from the pluck sound sample in section §IIIA and its spectrogram, below.

The nut piezo pickup showed somewhat greater variability resulting from repositioning – yet another reason to leave explicit measures of nut motion to a future investigation.

2. *spectrograms*

Listening allows the brain to sense how individual plucks evolve in time. Individual plucks on the different necks sound very similar for a given neck and possibly different for different necks. Spectrograms offer a graphic mathematical attempt to capture this. Spectrograms are usually presented without comment about how they were constructed. For the non-initiate, explanation would distract from the point being made. To the practiced, it is clear that an enormous number of choices have been made to bring out the features that the

presenter wishes to highlight.

For example, Fast Fourier Transforms (FFT's) need a window function shape and length to compute a spectrum. Spectrograms have additional variables that determine how the amplitude is mapped onto colors and intensity. Even in practice, the choices of display parameters fill out a huge, many-dimensional space. The presenter searches for choices that convey the desired message — to the extent that the message is genuinely contained in the data. However, when you listen to a sound, your brain automatically performs analyses that cannot be reproduced by linear mathematical processing. It simultaneously analyzes the signal on many different time scales, feeds intermediate results into the other parallel efforts, and makes choices that determine the final perception.[4]

Fig. 2 shows spectrograms of the twelve-pluck recording presented in §IIIA. Frequency is vertical on a linear scale, and time is horizontal. As with any practical spectrum decomposition, there is a trade off of frequency resolution against time resolution. The intensity is mapped onto color and brightness. Here, the range from bright white to dark blue was chosen to correspond to an interval of 60 dB. For reference, 60 dB is considered to be roughly the interval between the minimum threshold for hearing and normal, everyday sounds. Close inspection of intensity and duration reveal similarities between the members of a given pair and some differences between necks. The lower figure is exactly the same data, resolution, and color choice but restricted to below 6000 Hz.

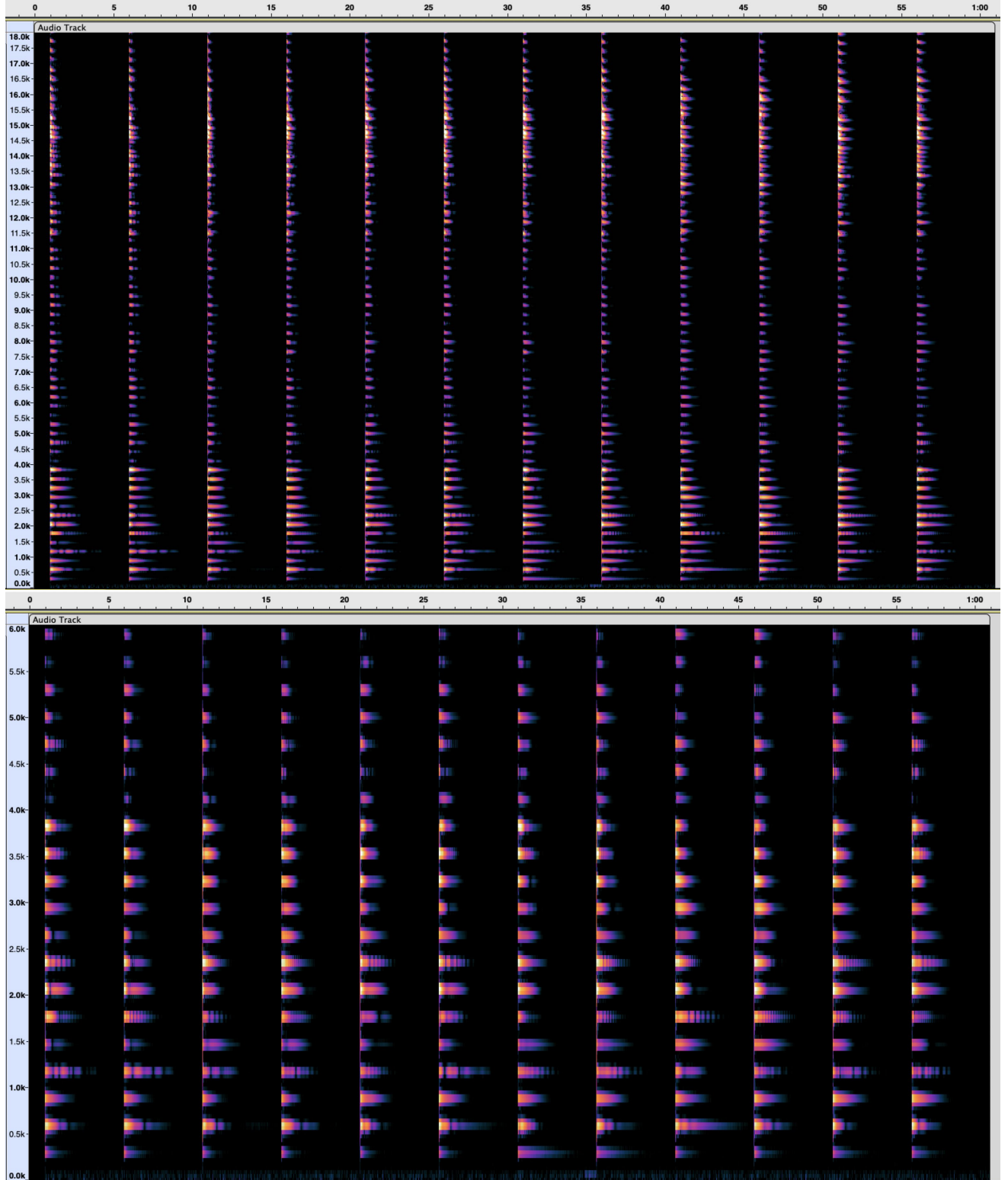


FIG. 2: Two plucks of maple I, maple II, mahogany I, mahogany II, walnut, & white oak

V. 1st AND 5th FRET CAPO

Each neck has its own glued-in nut. Differences among the nuts could potentially produce differences in the sounds of open strings. A capo at the 1st fret shifts the issue of the similarity of nuts to the similarity of the 1st frets. Fret wire is very uniform, and the remaining question is how well it and the next few frets are set.

With the capo moved to the 5th fret, it is reasonable to expect that the motion there due to the force of string vibration would be less than it is with the capo at the 1st fret because of the neck profile. Of course, any irregularity on one of the first four frets would cease to be relevant after capoing at the 5th.

Figs. 3, 4, 5, and 6 illustrate the result of three plucks with the maple neck on the left and three plucks with the mahogany neck on the right — first with the capo on the 1st fret and then with the capo on the 5th fret. Because the time scale is chosen to show the entire plucks, only the envelopes appear for each waveform. Audacity superimposes two curves. The dark blue shows the largest positive and negative peaks within the time interval corresponding to a single pixel in the chosen display time scale. The light blue is the RMS value for those time intervals.

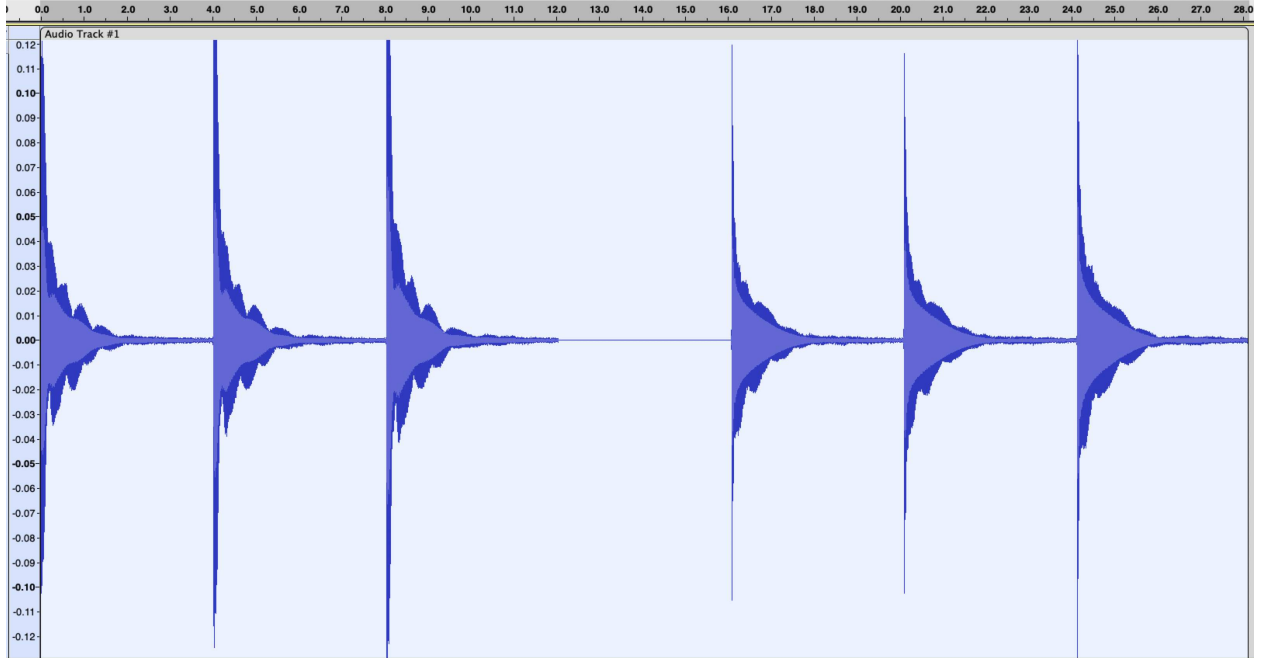


FIG. 3: 1st fret capo, 1st string plucks, maple (left) and mahogany (right)

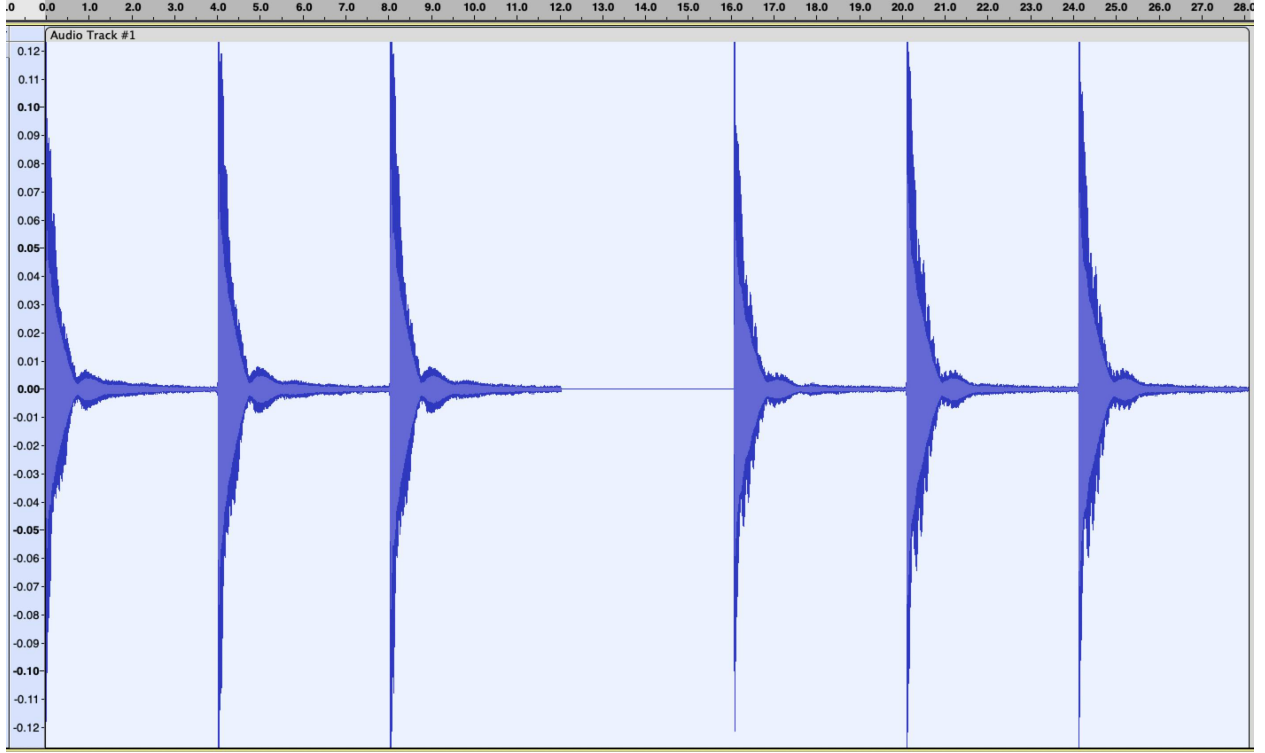


FIG. 4: 5th fret capo, 1st string plucks, maple (left) and mahogany (right)

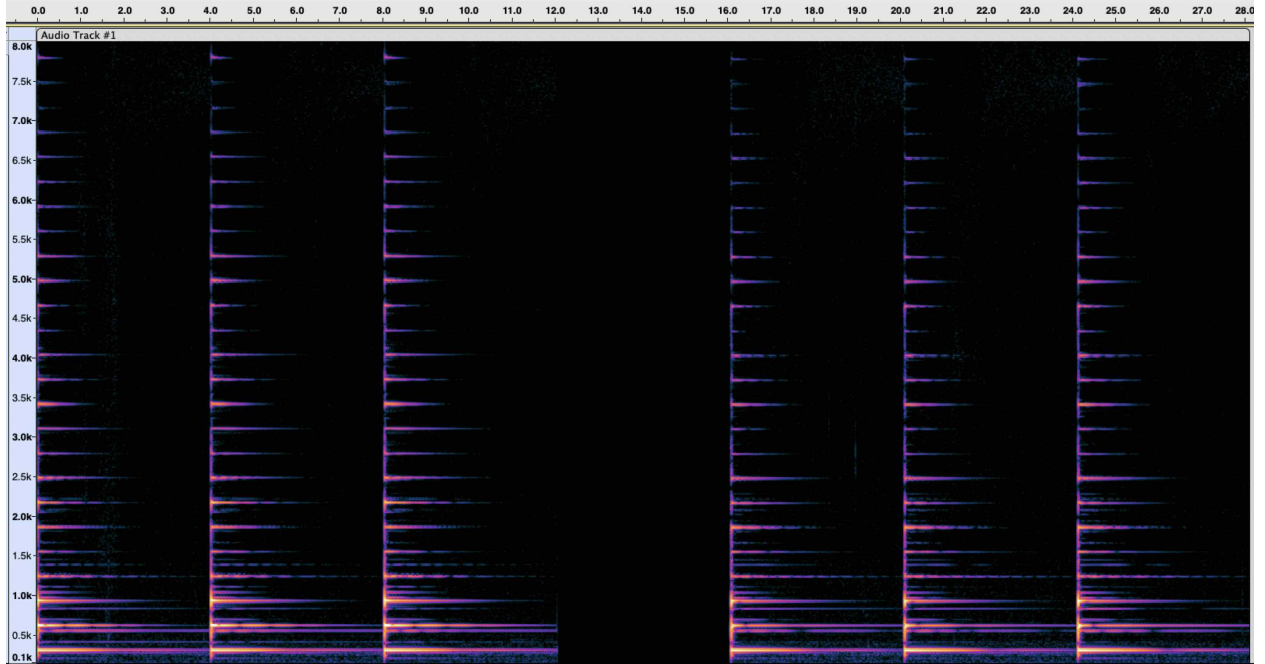


FIG. 5: 1st fret capo, 1st string plucks, maple (left) and mahogany (right)

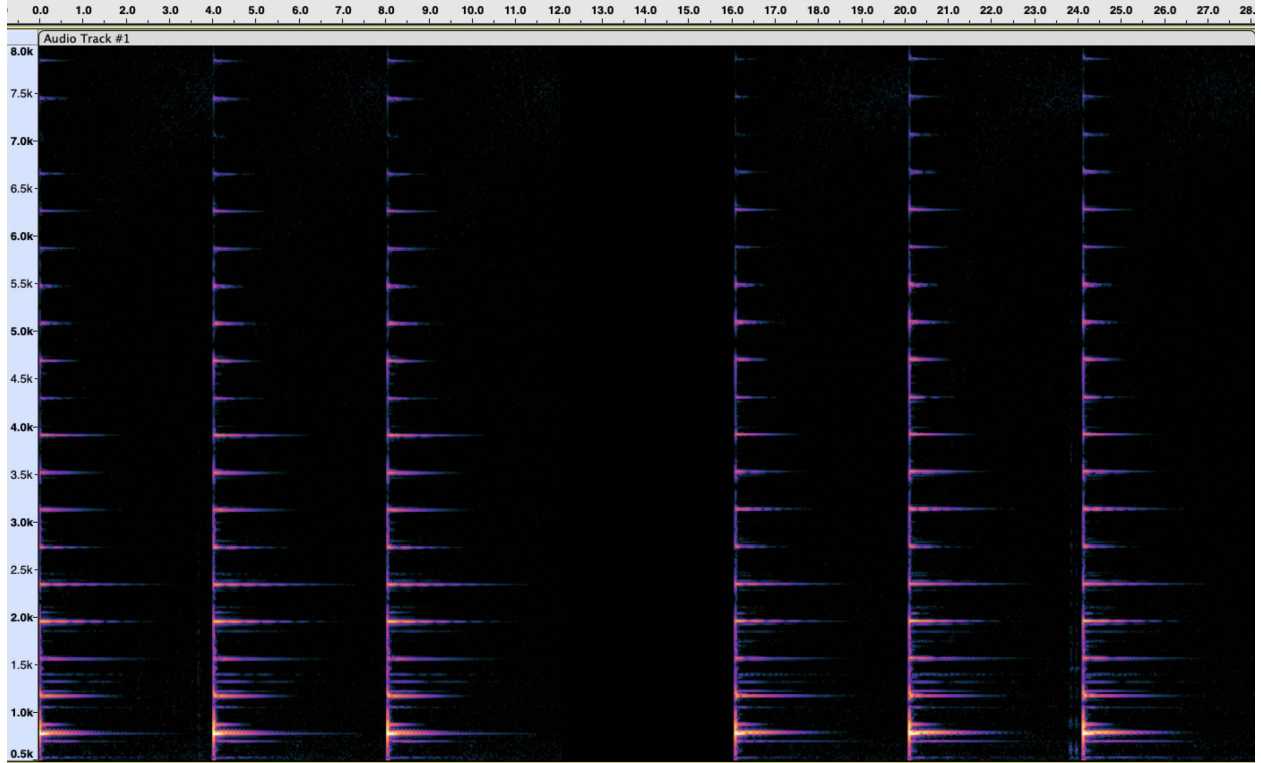


FIG. 6: 5th fret capo, 1st string plucks, maple (left) and mahogany (right)

Both the waveform envelopes and the spectrograms show distinct differences with the capo at the 1st fret and strong similarities with the capo at the 5th. This may simply be a reflection of there being less motion with the 5th fret capo.

VI. PHYSICAL SOUND SYNTHESIS TO ISOLATE THE NECK FROM THE REST

A different experiment was done on a Deering Eagle II, based on measurements of two-point admittance (i.e., at the bridge and at the nut) and the well-understood dynamics of the strings. These were combined to produce two versions of physical sound synthesis – one with the neck interacting with the rest of the banjo as determined from the measurements, the other with absolutely no neck vibration included in the calculation.

This is the sound at the bridge with the interacting, vibrating neck:

<http://www.its.caltech.edu/~politzer/neck-wood/Arkansas-traveller-NON-RIGID.mp3>

And this is the sound of the synthesized picking with an absolutely rigid neck:

<http://www.its.caltech.edu/~politzer/neck-wood/Arkansas-traveller-RIGID.mp3>

At the level of the discretization, the data in these files are not identical. However, as sounds to the ear, they certainly are. A little more explanation is in order.

If an oscillating force is applied to a system at a point, the ratio of resulting velocity of the system at that point to the applied force is the point admittance. This is naturally expressed as a complex function of frequency, whose phase represents the phase difference between the applied force and the resulting velocity. Admittance was measured at the Eagle II bridge for the study presented in ref. [6]. Physical sound synthesis uses the well-understood physics of strings combined with the measured admittance to predict the string and bridge motion after a pluck. The resulting bridge motion is presented as a surrogate for the produced sound. Those calculations assumed a perfectly rigid neck and include frequencies up to 6000 Hz. The methodology and appropriate references are given in ref. [6].

In the present situation, we want to include the neck motion at the nut in response to the force of the strings. Furthermore, a force at the nut produces motion at the bridge even if the strings are damped and cannot transmit any vibration themselves. This is easy to demonstrate with any banjo. Damp the strings and tap the nut. A substantial sound comes off the head. The reverse is also true (the two are directly related by a mathematical result called the “reciprocal theorem”). A tap on the bridge produces motion at the nut. These effects are represented by a two-point admittance, which is a complex 2×2 matrix function of frequency. It can be measured by measuring both motions in response to both types of taps. The admittance combines with the string dynamics analogously to what was done in ref. [6] to produce the motion of the bridge that results from a pluck.

One interpretation of this sound comparison is that the inevitable bending and twisting of the Eagle II maple neck, with its ebony fretboard and truss rod, are not sufficient to produce a sound that is distinguishable from a neck with no motion at all. Replacing the maple with mahogany would make the neck lighter and more flexible. In general, that would produce larger amplitude motion at the nut. The sound might change sufficiently to distinguish it from maple. In any case, more generally the added weight and stiffness of a typical, separate fretboard and truss rod can be expected to reduce the mechanical differences between necks of different timbers. The Goodtime necks in the rest of this study are solid wood, without separate fretboard or truss rod.

VII. GRAIN VARIATION

In an ideal log, i.e., long and straight with circular rings, there are three natural perpendicular directions at any point: the long way, radially, and azimuthally, i.e., tangent to the rings. The stiffness of the wood differs significantly for bending relative to those directions.[7] It is natural to choose lumber with grain going the long way to match the



FIG. 7: Neck end views

long way of the neck. The transverse grain is another matter. For violin luthiery, trees are milled to produce pieces with the grain oriented along the natural axes of the intended pieces. Banjo builders and certainly large-scale manufacturers likely do not usually have that luxury. Fig. 7 shows the ends of the four necks. Goodtime necks are pieced, with a separate heel block and a finger-jointed tuning head. So, it is the top sections in Fig. 7 that are most relevant to neck vibration. Identical wood will vibrate differently, depending on the orientation of the shaping relative to the grain and the rings. Of course, there is significant variation within a species, in addition to the variation due to milling geometry.

VIII. NECK PHYSICAL PARAMETERS

In any physics story to account for the neck impact on sound, the neck physical parameters would be essential to estimating magnitudes. No clear story is suggested here, but the parameters may be of some interest and potential future use. The various measurements are all collected in Fig. 9, below, at the end of this section. Numbers and ranges quoted by one of many comparable sources[8] for wood at 12% moisture content are also included there.

A. density

The wood of the new maple neck weighed 16.0 ounces. Although the volumes of the necks were not measured, the weight ratios can be compared to those quoted in other sources. The ratios would also be relevant in a comparison of simpler geometries, e.g., the resonances of rectangular bars.

The neck wood weight ratios are

$$\text{maple} : \text{mahogany} : \text{walnut} : \text{white oak} = 1 : 0.65 : 0.77 : 0.97 .$$

B. stiffness

The relative bending stiffness of the necks was estimated as follows. The necks were mounted on the rim, and the rim was firmly clamped, with the whole instrument horizontal. Weights were placed on the tuning head. The total added weight varied between 2 oz and 4 lbs. The downward deflection at the nut was measured.

Fig. 8 shows the measurements and least-squares fits. Maple was the stiffest. The displacement at the nut was about 7.7×10^{-5} inches/gram. The other necks were essentially all 11.0×10^{-5} inches/gram. Again, the ratios of these numbers are directly relevant to comparisons with other posted values.

No attempt was made to quantify torsional stiffness.

C. damping

The decay times of low lying flexural modes were estimated as follows. The unattached necks were supported in two places from below by rubber bands stretched across a frame.

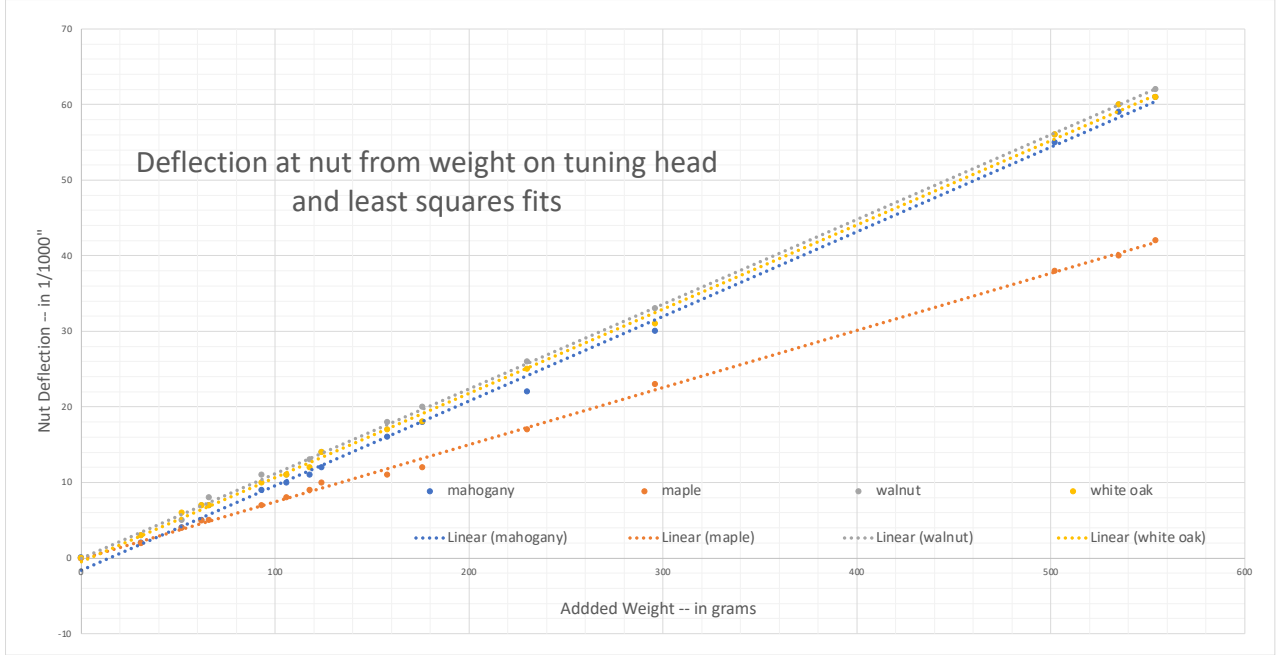


FIG. 8: Static neck bending

The sound of taps at an anti-node were recorded with a microphone near the tap and very near the neck surface. Each recording was processed through a narrow band-pass filter, and the resulting slope of the decay envelope was estimated.

The lowest resonant frequencies observed for that set-up for maple, mahogany, walnut, and white oak, respectively, were 147 Hz, 139 Hz, 126 Hz, and 114 Hz. As described, the decay rates were observed as decibels per second: 30, 22, 27, and 31 for the species in the same order. These imply Quality Factors, Q , equal to 67, 86, 63, and 51. These suggest that the mahogany resonances may generally have more sustain than the other woods. (Q is a dimensionless measure of the decay time in units of the oscillation period.) Note that others' very careful study of the decay of the sound produced by struck marimba bars suggests that the modal decays due to internal friction have a more complicated form.[9]

For systems that differ only by bulk density and stiffness, the frequencies are expected to be proportional to the square root of the ratio of stiffness to density. The frequencies relative to maple (in the same order, i.e., maple : mahogany : walnut : white oak) are 1 : 0.95 : 0.86 : 0.76 while the ratios of square root of stiffness-divided-by-total-mass are 1 : 0.97 : 0.91 : 0.84. So, for the lowest resonance, this little bit of physics is qualitatively correct and good to better than 10%.

As mentioned in the Introduction, the tuners together weigh nearly half as much as the mahogany. So, the total weight distribution differs significantly among the necks, and simple scaling of all higher frequency ratios is not expected.

NECK BULK PHYSICAL PROPERTIES

	maple	mahogany	walnut	white oak
total weight (incl. all hardware) - gm	607	448	503	595
wood (only) weight / maple	1	0.65	0.77	0.97
on-line density / maple	0.76 - 1.00	0.87	0.87	1.02 - 1.40
stiffness in gm/10 ⁻⁶ in	1.3	0.9	0.9	0.9
on-line Young's modulus / maple	0.63 - 1.00	0.69	0.92	0.56 - 1.09
1st resonance - Hz	147	139	126	114
1st resonance frequency / maple	1	0.95	0.86	0.78
sqrt(stiffness/total weight) / maple	1	0.97	0.91	0.84
Q-factor	67	86	63	51

FIG. 9: Neck bulk properties: measurements & on-line quotes from ref. [8]. The “on-line” ratios to maple shown above are relative to the high end of the quoted maple range, which is sometimes the only maple value quoted. The hardware common to all necks is 153 gm.

IX. PRIVILEGED POSITION OF THE PLAYER

The sound radiation pattern of the banjo (as with all other instruments) is highly frequency-dependent. Furthermore, sound in the “near field” close to the instrument can be very different from sound in the “far field”, many wavelengths away. For both reasons, the timbre perceived by the player may be rather different from that of a removed listener. On the subject of neck wood, some players sense distinct differences between the species. Their hands may play a role in that, not just their ears.

Some people can certainly feel the vibrations of the neck. The deaf percussionist Evelyn Glennie[10] hears with her hands, feet, and her whole body. For those with normal hearing, it is unknown to what extent such stimuli might impact their perception of sound.

The player also has a myriad of ways to alter the sound of a particular note. The effects can be far greater than the differences observed here for the same sort of pluck on different necks. The player is aware (consciously or not) of how a given pluck was performed and

then hears the result. The different species necks change the relation between those two, even if only slightly. But the change might be enough to register in terms of perception. On the other hand, the listener (live or recorded) has no idea of the details of the player’s mechanics but only judges by the sound. It may be more difficult to separate the species dependence without knowledge of the player’s technique.

X. COMPLEXITY OF THE PHYSICS

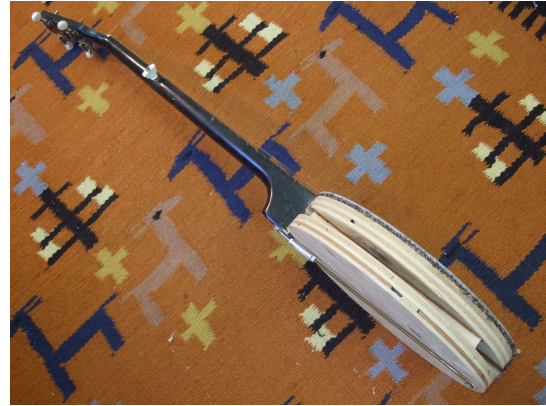
Imagine an instrument where the neck was connected to the rest of the banjo solely by the strings. That would require a heel joint that prevented any vibration from being transferred between neck and rim. Given that the neck is a very inefficient radiator of sound, neck motion resulting from the force of strings at the nut would be dissipated as heat in the wood. In this imagined situation, more motion at the nut leads to less produced sound from the head.

However, as mentioned in section §VI, if you tap on the nut with strings damped or absent, plenty of sound comes off the head. Neck vibrations are transferred at the heel to the rim and *vice versa*. Energy can flow in either direction. The normal modes of an unmounted neck (like any other “bending beam” structure) will get further and further apart with increasing frequency. However, there are many in the acoustic range. The head modes behave differently as frequency increases: they become dense above the lowest few. The complexities of the data and this feature of the construction suggest that no simple modeling will capture the frequency dependence of neck sound and relate it to bulk parameters. However, it is an open challenge.

XI. SUMMARY

Sound differences between necks, possibly attributable to physical properties of the different wood species, were identified. They are small but discernible. They are also difficult to characterize.

- [1] Touch the tuning head of a solid-body guitar or banjo (e.g., shown) on a large thin surface, e.g., cabinet door, thin table top, door skin of an inside door. A substantial sound of a pluck will come off that surface. If you put your ear close, you don't even need the door or table top.



- [2] These 3 second clips are also available at <https://euphonics.org>, section 5.5.4.
- [3] <http://www.its.caltech.edu/~politzer/neck-wood/key-to-four-necks.pdf>, HDP: 22 – 02a, *Wood Identification Key for Four Necks*
- [4] recommended: two recent books by leading scientists for the interested layman about sound, music, and the brain: N. Krauss, *Of Sound Mind – How Our Brain Constructs a Meaningful Sonic World*, MIT Press (2021); and D. Deutsch, *Musical Illusions and Phantom Words: How Music and Speech Unlock Mysteries of the Brain*, Oxford University Press (2019)
- [5] D. Politzer, *The Open Back of the Open-Back Banjo*, HDP: 13 – 02, <http://www.its.caltech.edu/~politzer>, December 2013
- [6] a technical exposition is available as open-access as *Acoustics of the Banjo: measurements and sound synthesis & theoretical and numerical modeling*, J. Woodhouse, D. Politzer, and H. Mansour, *Acta Acustica*, **5**, 15 and 16 (2021) <https://doi.org/10.1051/aacus/2021009> and <https://doi.org/10.1051/aacus/2021008>; *Pickers' Guide to Acoustics of the Banjo*, D. Politzer, J. Woodhouse, and H. Mansour, HDP: 21 – 01, <http://www.its.caltech.edu/~politzer> – APRIL 2021 is an informal account of some of the salient results.
- [7] e.g., <https://www.fpl.fs.fed.us/documnts/pdf2001/green01d.pdf>
- [8] https://www.engineeringtoolbox.com/timber-mechanical-properties-d_1789.html
- [9] S. McAdams, A. Chaigne, & V. Roussarie, *JASA* **115** #3, 1306 (2004)

- [10] find interviews, video, &c of Evelyn Glennie on-line. For example:
at Grand Central Station (NYC): <https://www.youtube.com/watch?v=BHrE0QRcPIY> and
“How to Listen”: <https://archive.org/details/EvelynGlennie.2003>